Ordinal addition is associative

The Open Logic Project contributors

openlogic:set-theory/ord-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ord-arithmetic/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted clause is ordaddition3 only; the author's own 'leaving the rest as an exercise' covers clauses 1/2/4 which are not part of this card

Editorial note. The label ordaddition3 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Ordinal addition recursion equations. Same unit, \ollabel{ordadditionrecursion} (statement lines 489-496). Supplies the successor step (alpha+beta)+1 = alpha+(beta+1) shape and the limit supremum characterisation used in all three induction cases.

Source reference. Original lemma ordadditionrecursion: sources/olp-set-theory-ord-arithmetic/pinned-original-source.tex, source lines 489–496. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If α, β, γ are ordinals, then:*

1. *if $\beta < \gamma$, then $\alpha + \beta < \alpha + \gamma$*
2. *if $\alpha + \beta = \alpha + \gamma$, then $\beta = \gamma$*
3. *$\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$, i.e., addition is associative*
4. *If $\alpha \leq \beta$, then $\alpha + \gamma \leq \beta + \gamma$*

Proof. We prove 3, leaving the rest as an exercise. The proof is by Simple Transfinite Induction on γ, using ordadditionrecursion. When $\gamma = 0$:

$$(\alpha + \beta) + 0 = \alpha + \beta = \alpha + (\beta + 0)$$

When $\gamma = \delta + 1$, suppose for induction that $(\alpha + \beta) + \delta = \alpha + (\beta + \delta)$; now using ordadditionrecursion three times:

$$\begin{aligned}(\alpha + \beta) + (\delta + 1) &= ((\alpha + \beta) + \delta) + 1 \\ &= (\alpha + (\beta + \delta)) + 1 \\ &= \alpha + ((\beta + \delta) + 1) \\ &= \alpha + (\beta + (\delta + 1))\end{aligned}$$

When γ is a limit ordinal, suppose for induction that if $\delta \in \gamma$ then $(\alpha+\beta)+\delta = \alpha + (\beta + \delta)$; now:

$$\begin{aligned}(\alpha + \beta) + \gamma =& \subsetneq_{\delta<\gamma} ((\alpha + \beta) + \delta) \\ =& \subsetneq_{\delta<\gamma} (\alpha + (\beta + \delta)) \\ =& \alpha + \subsetneq_{\delta<\gamma} (\beta + \delta) \\ =& \alpha + (\beta + \gamma)\end{aligned}$$

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